Week 1 Exercise Solutions

1.

For $c \in \mathbb{R}$, let

$$S_c = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = c\}.$$

Then S_c is a subspace of $\mathbb{R}^3$ if and only if $c = 0$.

Proof. If S_c is a subspace, it contains $(0,0,0)$. Substituting this vector into the defining equation gives $c = 0$.

Conversely, suppose $c = 0$. The zero vector belongs to S_0. Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ belong to S_0, and let $\alpha, \beta \in \mathbb{R}$. Then

$$\begin{aligned}(\alpha x_1 + \beta x_2) + (\alpha y_1 + \beta y_2) - (\alpha z_1 + \beta z_2) \\ = \alpha(x_1 + y_1 - z_1) + \beta(x_2 + y_2 - z_2) = 0.\end{aligned}$$

Thus $\alpha u + \beta v \in S_0$. By the subspace criterion, S_0 is a subspace. ▫

2.

Let

$$U = \{(a, b, 0) : a, b \in \mathbb{R}\}, \qquad W = \{(0, c, d) : c, d \in \mathbb{R}\}.$$

We have

$$U \cap W = \{(0, t, 0) : t \in \mathbb{R}\},$$

and the sum $U + W$ is not direct.

Proof. A vector in $U \cap W$ has third coordinate zero because it belongs to U, and first coordinate zero because it belongs to W. It therefore has the form $(0, t, 0)$. Conversely, every vector of this form belongs to both U and W. Therefore,

$$U \cap W = \{(0, t, 0) : t \in \mathbb{R}\}.$$

If $U + W$ is direct, the only intersection vector need to be $(0,0,0)$, but it fails. Therefore, the sum is not direct.▫

3.

Let

$$P = \{p \in \mathbb{R}[x] : p(1) = 0\}.$$

Then P is a subspace of $\mathbb{R}[x]$.

Proof. The zero polynomial satisfies $0(1) = 0$, so $0 \in P$. Let $p, q \in P$ and $\alpha, \beta \in \mathbb{R}$. The function $\alpha p + \beta q$ is a real polynomial, and

$$(\alpha p + \beta q)(1) = \alpha p(1) + \beta q(1) = \alpha \cdot 0 + \beta \cdot 0 = 0.$$

Therefore $\alpha p + \beta q \in P$. The subspace criterion now shows that P is a subspace of $\mathbb{R}[x]$. ▫

4.

The solution set is the zero subspace over $\mathbb{R}$. However, it's not a subspace over $\mathbb{C}^2$.

Proof. If $x, y \in \mathbb{R}$, then $x^2 \geq 0$ and $y^2 \geq 0$. Consequently,

$$x^2 + y^2 = 0 \quad \Leftrightarrow \quad x = y = 0.$$

Thus the real solution set is $\{(0,0)\}$, which is a subspace of $\mathbb{R}^2$.

For complex coordinates, factor the defining expression:

$$x^2 + y^2 = (y - ix)(y + ix).$$

Since $\mathbb{C}$ has no zero divisors, the product is zero exactly when $y = ix$ or $y = -ix$. Hence the complex solution set is

$$\{(z, iz) : z \in \mathbb{C}\} \cup \{(z, -iz) : z \in \mathbb{C}\}.$$

In particular, $(1, i)$ and $(1, -i)$ both belong to it. Their sum is $(2,0)$, which does not, because its coordinates satisfy $2^2 + 0^2 = 4 \neq 0$. The set is not closed under addition and therefore is not a subspace. ▫

5.

Let U be a subspace of a real vector space V, and let $v_0 \in V$. Define

$$v_0 + U = \{v_0 + u : u \in U\}.$$

We prove that $v_0 + U$ contains the entire line through any two of its distinct points.

Proof. Let $x, y \in v_0 + U$ with $x \neq y$. There exist $u, w \in U$ such that $x = v_0 + u$ and $y = v_0 + w$. The line through x and y is

$$L = \{x + t(y - x) : t \in \mathbb{R}\}.$$

For any $t \in \mathbb{R}$,

$$x + t(y - x) = (v_0 + u) + t(w - u) = v_0 + (1 - t)u + tw.$$

Because U is a subspace, $(1 - t)u + tw \in U$. Thus $x + t(y - x) \in v_0 + U$ for every real t, proving $L \subseteq v_0 + U$. ▫

The parameter ranges over all of $\mathbb{R}$, not merely $[0,1]$, so the argument proves containment of the whole line, including the points beyond x and y.

If V is a complex vector space and U is a complex subspace, the same calculation works for every $t \in \mathbb{C}$. It proves containment of the complex affine line; taking only real t also proves the real-line statement.